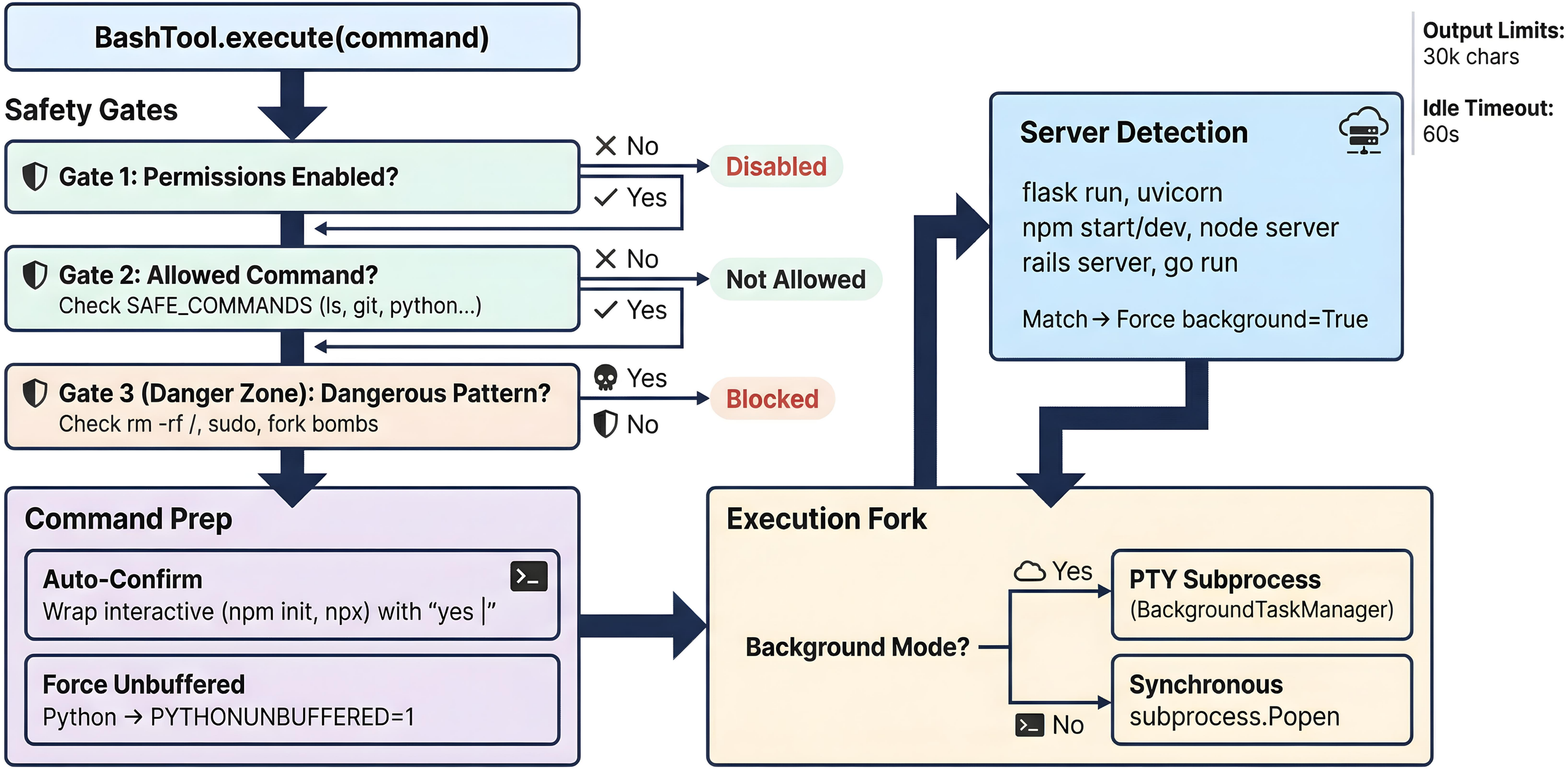
BashTool.execute(command)
Safety Gates
Gate 1: Permissions Enabled?
No
Disabled
Yes
Gate 2: Allowed Command?
Check SAFE_COMMANDS (ls, git, python...)
No
Not Allowed
Yes
Gate 3 (Danger Zone): Dangerous Pattern?
Check rm -rf /, sudo, fork bombs
Yes
Blocked
No
Command Prep
Auto-Confirm
Wrap interactive (npm init, npx) with "yes |"
Force Unbuffered
Python → PYTHONUNBUFFERED=1
Execution Fork
Background Mode?
Yes
PTY Subprocess
(BackgroundTaskManager)
No
Synchronous
subprocess.Popen
Server Detection
flask run, uvicorn
npm start/dev, node server
rails server, go run
Match → Force background=True
Output Limits:
30k chars
Idle Timeout:
60s